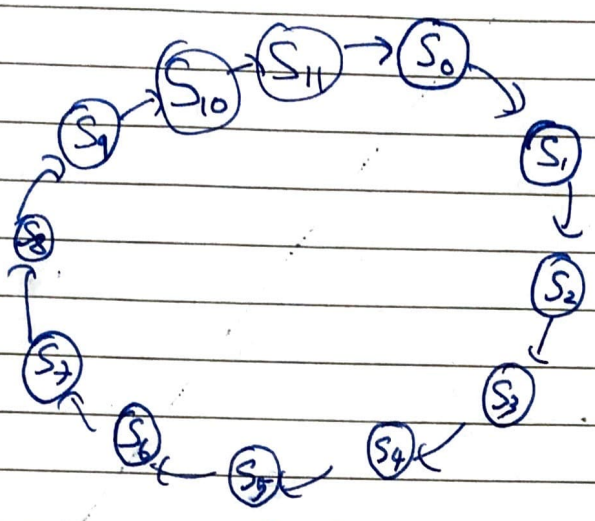
Aayush Kumar 24095931B
71 Aayush Date :

1 Mod 12 synchronous counter

(i) using JK ff

State Diagram



State table

PS	NS
S_0	S_1
S_1	S_2
S_2	S_3
S_3	S_4
S_4	S_5
S_5	S_6
S_6	S_7
S_7	S_8
S_8	S_9
S_9	S_{10}
S_{10}	S_{11}
S_{11}	S_0

$Q_3\ Q_2\ Q_1\ Q_0 \quad Q_3^+\ Q_2^+\ Q_1^+\ Q_0^+$

State Assignment

PS	NS
0 0 0 0	0 0 0 1
0 0 0 1	0 0 1 0
0 0 1 0	0 0 1 1
0 0 1 1	0 1 0 0
0 1 0 0	0 1 0 1
0 1 0 1	0 1 1 0
0 1 1 0	0 1 1 1
0 1 1 1	1 0 0 0
1 0 0 0	1 0 0 1
1 0 0 1	1 0 1 0
1 0 1 0	1 0 1 1
1 0 1 1	0 0 0 0
1 1 0 0	x x x x
1 1 0 1	x x x x
1 1 1 0	x x x x
1 1 1 1	x x x x

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State excitation

PS $Q_3\ Q_2\ Q_1\ Q_0$	NS $Q_3^+\ Q_2^+\ Q_1^+\ Q_0^+$	$J_3\ K_3\ J_2\ K_2\ J_1\ K_1\ J_0\ K_0$
0 0 0 0	0 0 0 1	0X 0X 0X 1X
0 0 0 1	0 0 1 0	0X 0X 1X X1
0 0 1 0	0 0 1 1	0X 0X X0 1X
0 0 1 1	0 1 0 0	0X 1X X1 X1
0 1 0 0	0 1 0 1	0X X0 0X 1X
0 1 0 1	0 1 1 0	0X X0 1X X1
0 1 1 0	0 1 1 1	0X X0 X0 1X
0 1 1 1	1 0 0 0	1X X1 X1 X1
1 0 0 0	1 0 0 1	X0 0X 0X 1X
1 0 0 1	1 0 1 0	X0 0X 1X X1
1 0 1 0	1 0 1 1	X0 0X X0 1X
1 0 1 1	0 0 0 0	X1 0X X1 X1

$J_0 = 1$
$K_0 = 1$

J_0 — Q_3Q_2 / Q_1Q_0

	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	X	X	X	X
10	1	X	X	1

K_0 — Q_3Q_2 / Q_1Q_0

	00	01	11	10
00	X	1	1	X
01	X	1	1	X
11	X	X	X	X
10	X	1	1	X

$J_1 = Q_0$
$K_1 = Q_0$

J_1 — Q_3Q_2 / Q_1Q_0

	00	01	11	10
00	0	1	X	X
01	0	1	X	X
11	X	X	X	X
10	0	1	X	X

K_1 — Q_3Q_2 / Q_1Q_0

	00	01	11	10
00	X	X	1	0
01	X	X	1	0
11	X	X	X	X
10	X	X	1	0

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$\bar{J}_2$

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	X	X	X	X
11	X	X	X	X
00	0	0	0	0

$\bar{J}_2 = Q_0Q_1Q_3$

$K_2 = Q_0Q_1$

K_2

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	X	X	X	X
01	0	0	1	0
11	X	X	X	X
10	X	X	X	X

$\bar{J}_3$

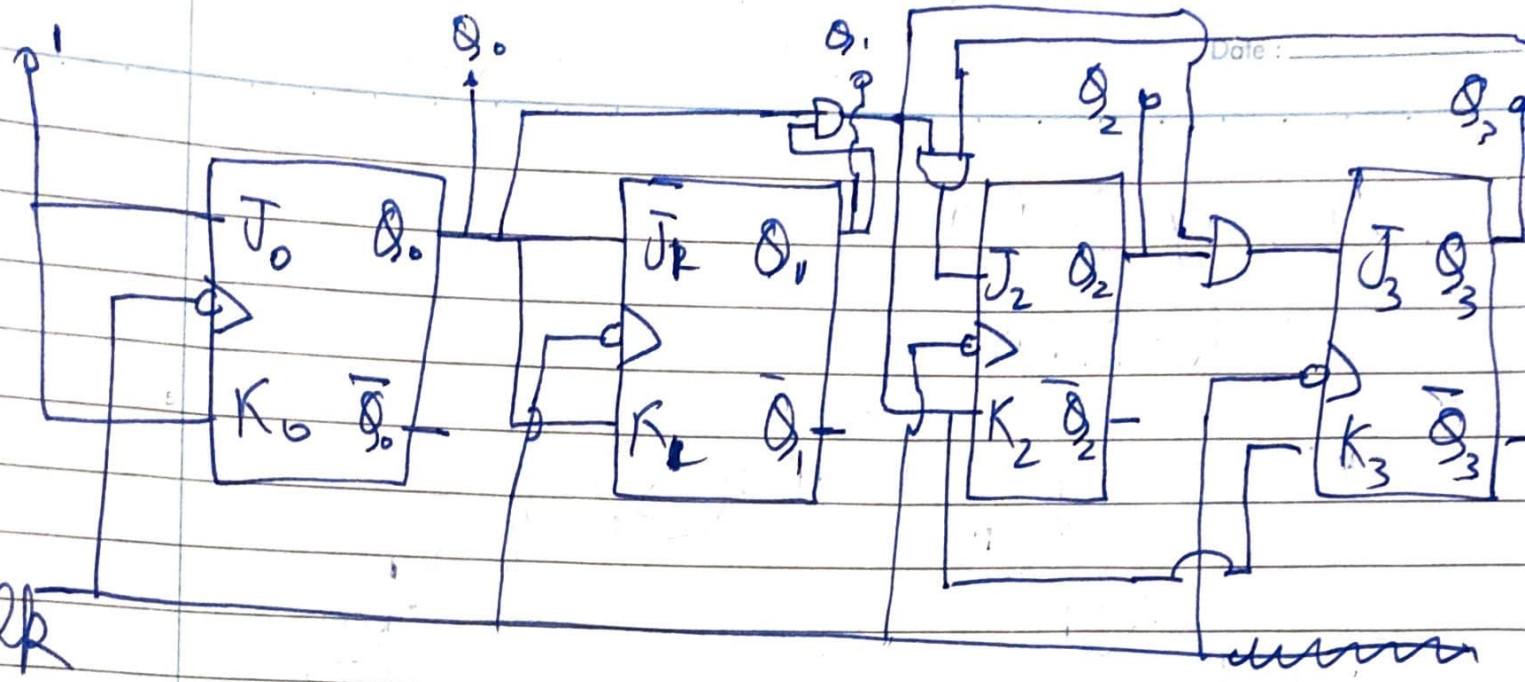
Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	X	X	X	X
10	X	X	X	X

$\bar{J}_3 = Q_2Q_1Q_0$

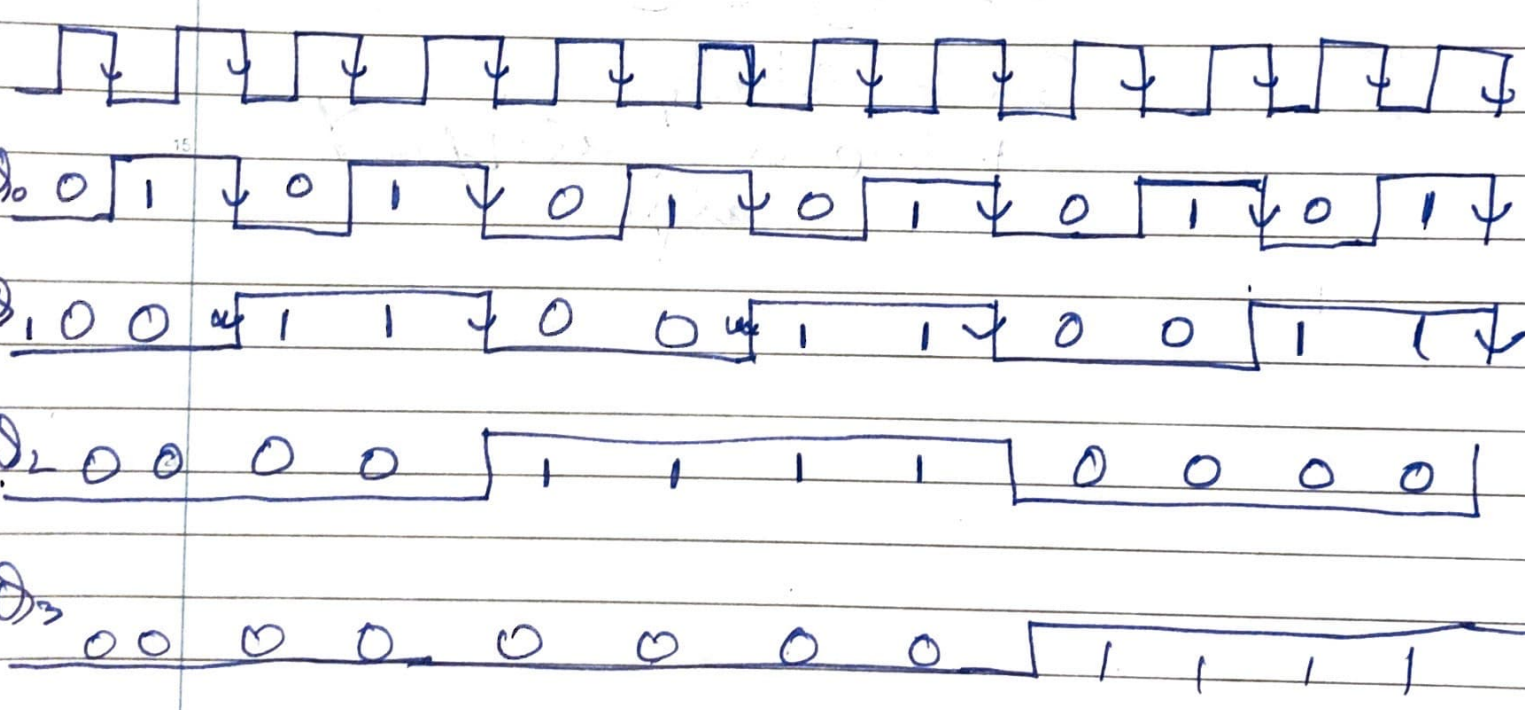
$K_3 = Q_1Q_0$

K_3

Q_3Q_2 \ Q_1Q_0	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	X	X	X	X
10	0	0	1	0



Timing diagram

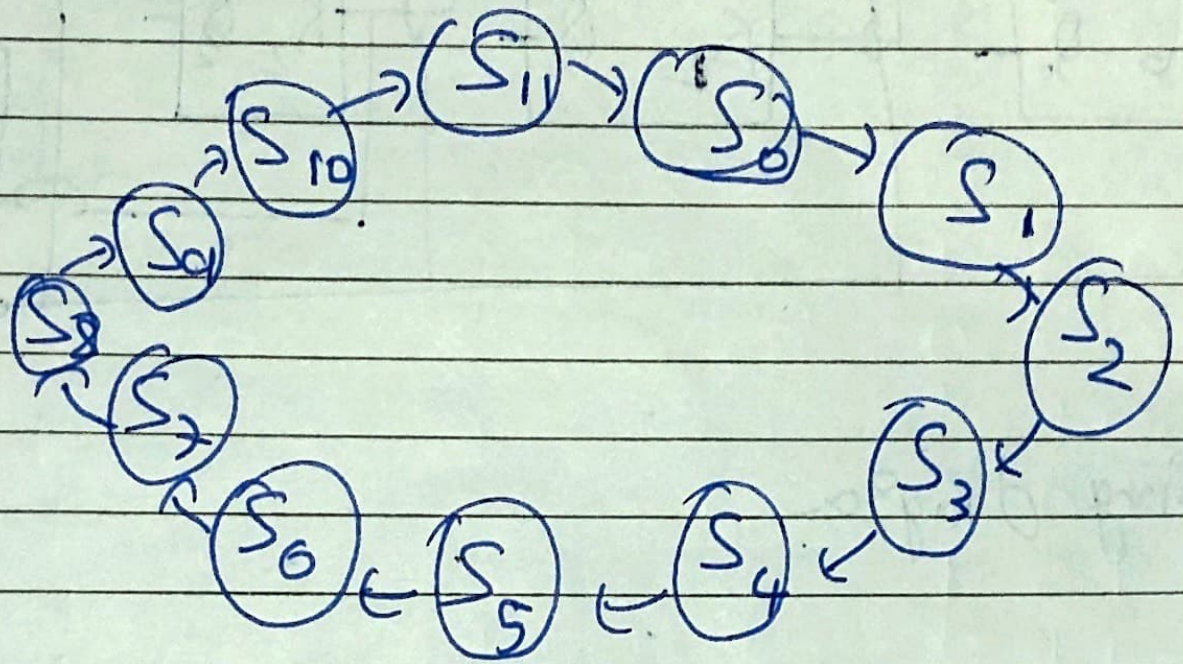


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(ii) Using D ff

State Diagram



Same state table and assigned as (i)

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State Excitation

PS				NS							
Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	0
0	0	1	0	0	0	1	1	0	0	1	1
0	0	1	1	0	1	0	0	0	1	0	0
0	1	0	0	0	1	0	1	0	1	0	1
0	1	0	1	0	1	1	0	0	1	1	0
0	1	1	0	0	1	1	1	0	1	1	1
0	1	1	1	1	0	0	0	1	0	0	0
1	0	0	0	1	0	0	1	1	0	0	1
1	0	0	1	1	0	1	0	1	0	1	0
1	0	1	0	1	0	1	1	1	0	1	1
1	0	1	1	0	0	0	0	0	0	0	0

D_0 — Q_1Q_0 / Q_3Q_2

	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	x	x	x	x
10	1	0	0	1

D_1

	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	x	x	x	x
10	0	1	0	1

D_2

	00	01	11	10
00	0	0	1	0
01	1	1	0	1
11	x	x	x	x
10	0	0	0	0

D_3

	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	x	x	x	x
10	1	1	0	1

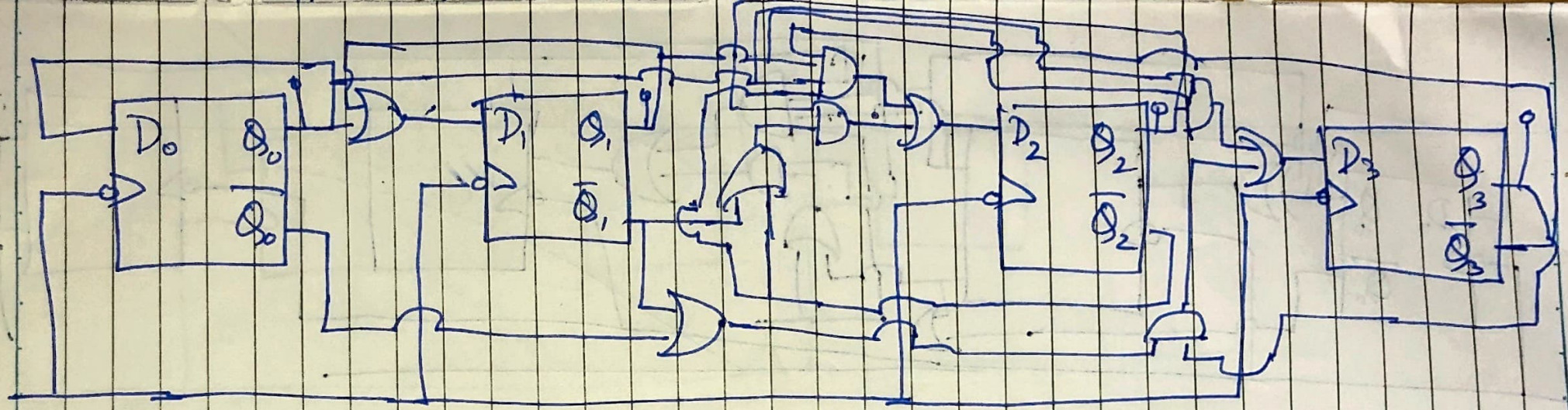
$$D_0 = \overline{Q_0}$$

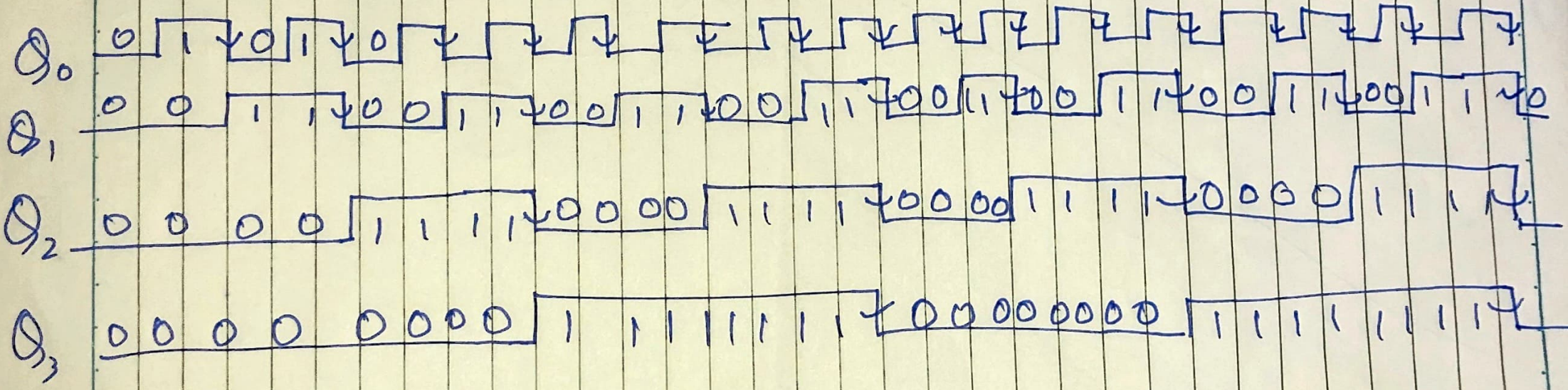
$$D_1 = \overline{Q_1}Q_0 + Q_1\overline{Q_0}$$

$$D_2 = \overline{Q_3}\,\overline{Q_2}Q_1Q_0 + Q_2\overline{Q_1} + Q_2\overline{Q_0}$$

$$D_3 = Q_3\overline{Q_1} + Q_3\overline{Q_0} + Q_2Q_1Q_0$$

Date: _____


D_0
Q_0
$\overline{Q_0}$
D_1
Q_1
$\overline{Q_1}$
D_2
Q_2
$\overline{Q_2}$
D_3
Q_3
$\overline{Q_3}$


Q_0
Q_1
Q_2
Q_3

Comparative analysis

Date : _____

	JK	D
Complexity	Simple requires fewer logic elements	Complex requires more logic elements
Hardware	4 ff & 3 gates	4 ff & 9 gates
Perfomance	Faster due to less gates	Slower due to delay of several gates

Even though JK is more efficient & requires lesser hardware D ff is easier to understand and debug due to its straight forward input output relation!

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2 | 0 0 0 1 1 1 0 | 1 0 0 0 1 1 1 0 1 0

(1) Direct

$$Max(0s, 1s) \leq 2^{n-1}$$

$$4 \leq 2^{n-1}$$

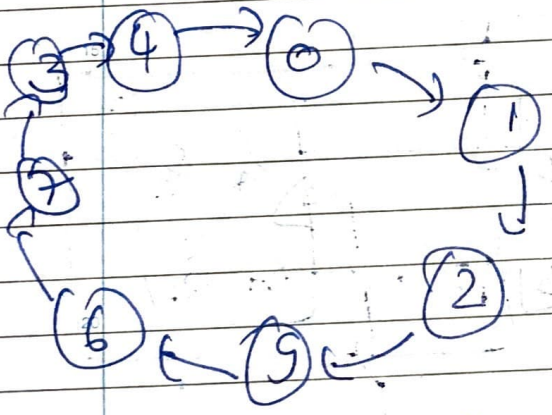
$$2^2 \leq 2^{n-1}$$

$$n-1 = 2$$

$$n = 3$$

Q_A	Q_B	Q_C	
1	0	0	4
0	0	0	0
0	0	1	1
0	1	0	2
1	0	1	5
1	1	0	6
1	1	1	7
0	1	1	3

State diagram



	Q_A	Q_B	Q_C	Q_A^+	Q_B^+	Q_C^+	J_A	K_A	J_B	K_B	J_C	K_C
4	1	0	0	0	0	0	X	1	0	X	0	X
0	0	0	0	0	0	1	0	X	0	X	1	X
1	0	0	1	0	1	0	0	X	1	X	X	1
2	0	1	0	1	0	1	1	X	X	1	1	X
5	1	0	1	1	1	0	X	0	1	X	X	1
6	1	1	0	1	1	1	X	0	X	0	1	X
7	1	1	1	0	1	1	X	1	X	0	X	0
3	0	1	1	1	0	0	1	X	X	1	X	1

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Q_A \ Q_BQ_C	00	01	11	10
0	0	0	1	1
1	X	X	X	X

$J_A = Q_B$

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	X	X
1	1	0	1	0

$K_A = \bar{Q}_B\bar{Q}_C + Q_BQ_C$

Q_A \ Q_BQ_C	00	01	11	10
0	0	1	X	X
0	1	X	X	

$J_B = Q_C$

Q_A \ Q_BQ_C	00	01	11	10
0	X	X	1	1
1	X	X	0	0

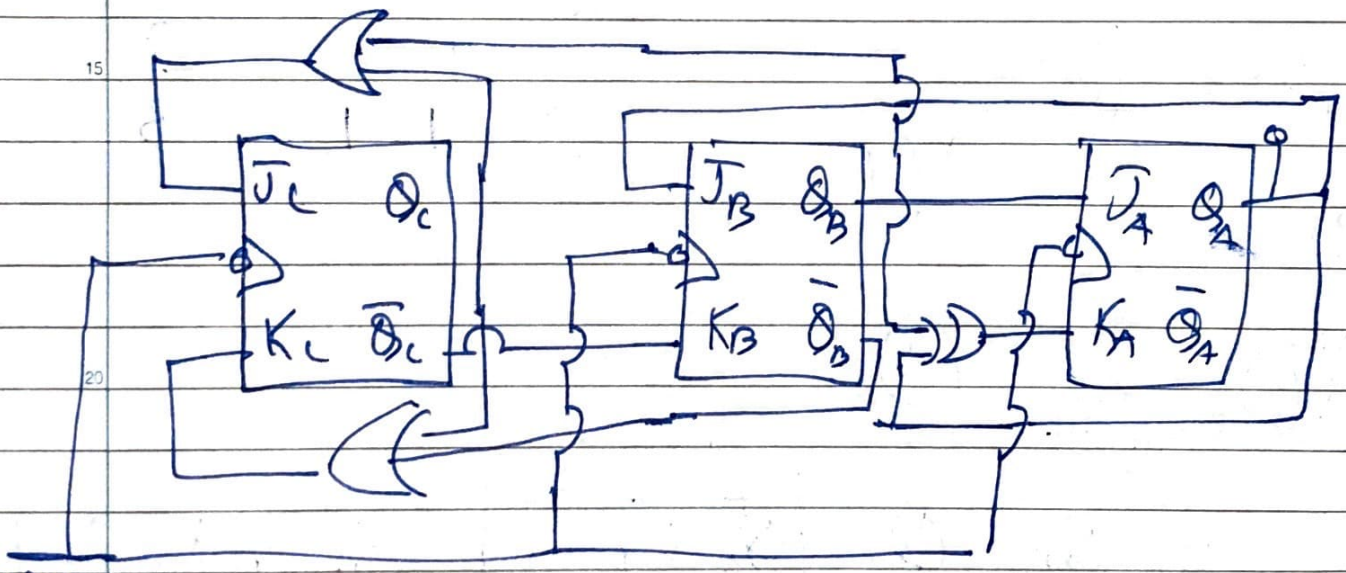
$K_B = \bar{Q}_A$

Q_A \ Q_BQ_C	00	01	11	10
1	X	X	1	1
0	X	X	1	1

$J_C = \bar{Q}_A + Q_B$

Q_A \ Q_BQ_C	00	01	11	10
1	X	1	1	X
X	1	0	X	

$K_C = \bar{Q}_A + \bar{Q}_B$



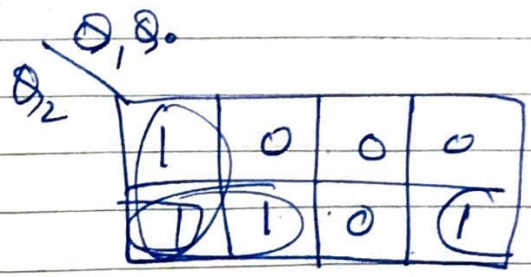
clk

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(11) indirect

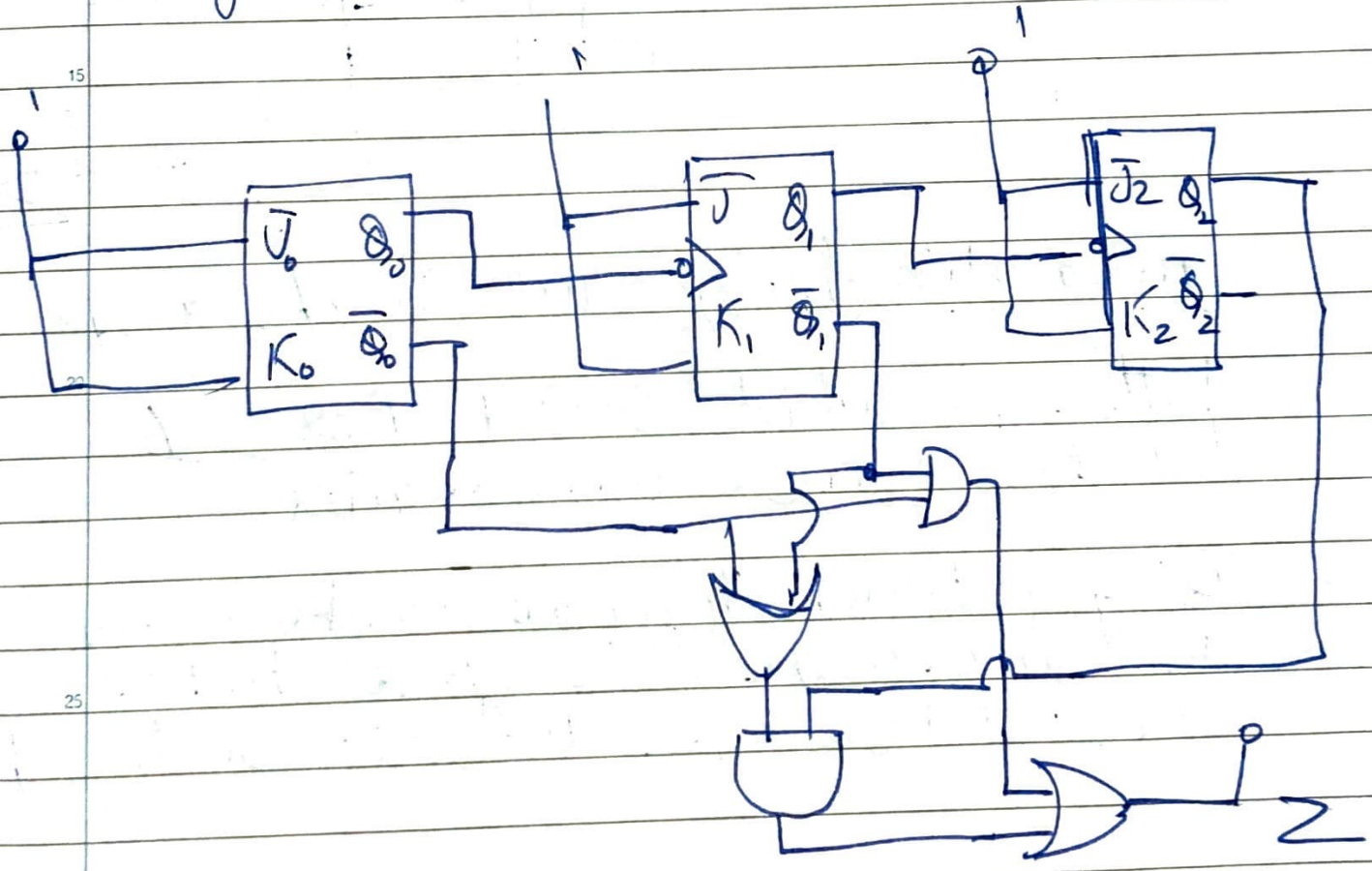
	Q_2	Q_1	Q_0	O/P
S_0	0	0	0	1
S_1	0	0	1	0
S_2	0	1	0	0
S_3	0	1	1	0
S_4	1	0	0	1
S_5	1	0	1	1
S_6	1	1	0	1
S_7	1	1	1	0



$$F = \bar{Q}_1\bar{Q}_0 + Q_2\bar{Q}_1 + Q_2\bar{Q}_0$$

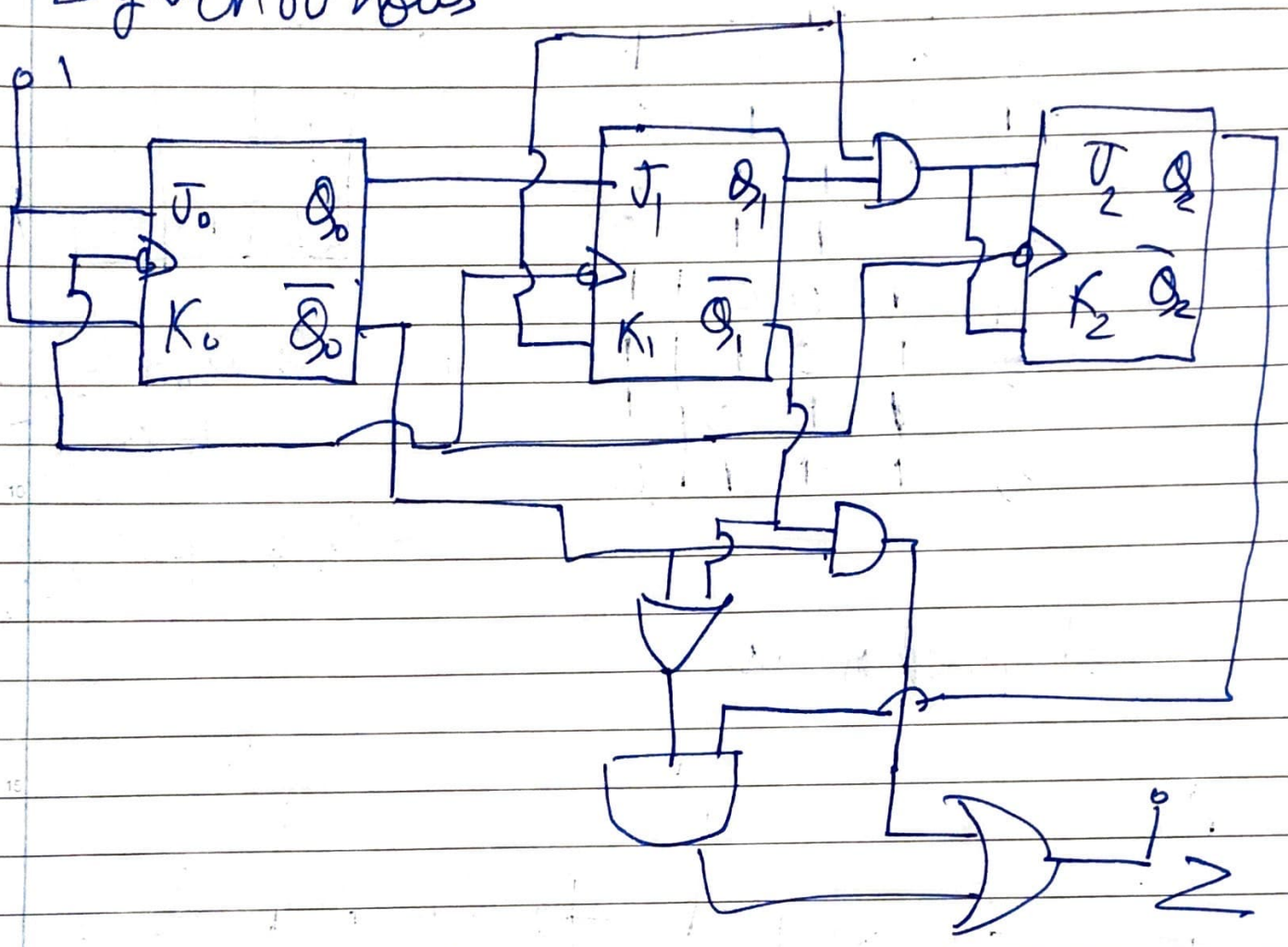
$$= \bar{Q}_1\bar{Q}_0 + Q_2(\bar{Q}_1 + \bar{Q}_0)$$

Asynchronous



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Synchronous



Using direct logic we make a synchronous pattern in which our MSB or LSB is the pattern we require

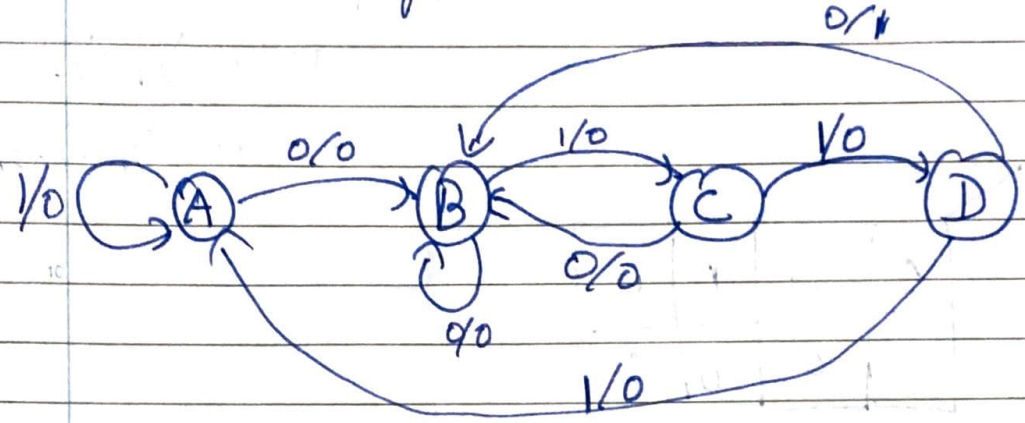
Using indirect logic we make a combinational ckt whose output corresponds to the req. pattern for different counter inputs.

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i) Overlapping

State Diagram



State table

PS	NS		O/P	
	x=0	x=1	x=0	x=1
A	B	A	0	0
B	B	C	0	0
C	B	D	0	0
D	B	A	1	0

State Assigned

PS	NS		O/P		state excitation	
	x=0	x=1	x=0	x=1	$Q_1 Q_0$ x=0	$Q_1 Q_0$ x=1
00	01	00	0	0	01	00
01	01	10	0	0	01	10
10	01	11	0	0	01	11
11	01	00	1	0	01	00

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D_0

Q_1Q_0 / w	00	01	11	10
0	1	1	1	1
1	0	0	0	1

$D_0 = \bar{w} + Q_1\bar{Q}_0$

D_1

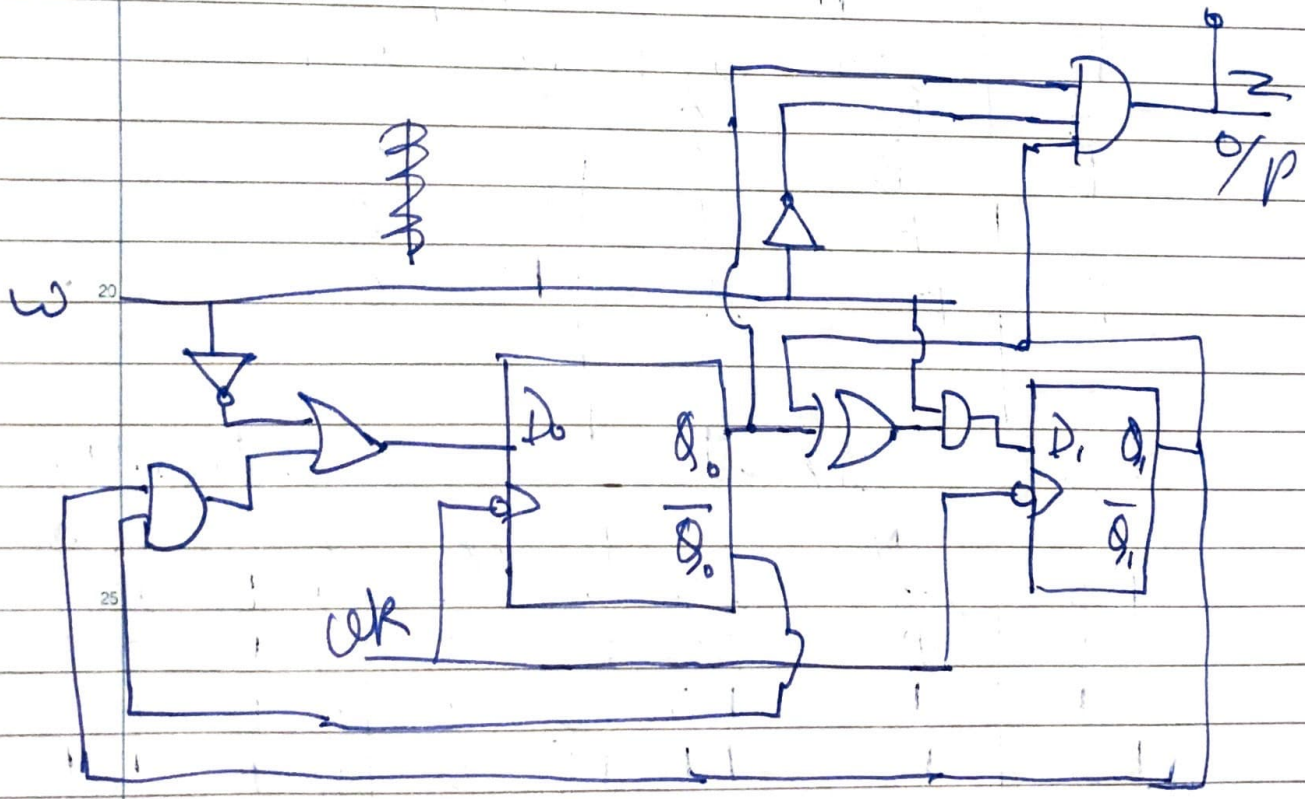
Q_1Q_0 / w	00	01	11	10
0	0	0	0	0
1	0	1	0	1

$D_1 = wQ_1Q_0 + w\bar{Q}_1\bar{Q}_0 = w(Q_1 \oplus Q_0)$

Z

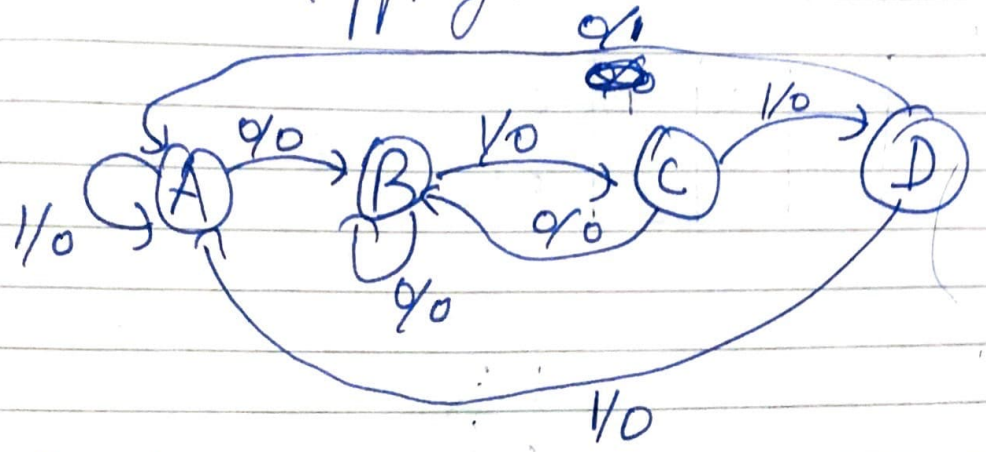
Q_1Q_0 / w	00	01	11	10
0	0	0	1	0
1	0	0	0	0

$Z = \bar{w}Q_1Q_0$



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(ii) Non Overlapping



State table

PS	NS		O/P	
	x=0	x=1	x=0	x=1
A	B	A	0	0
B	B	C	0	0
C	B	D	0	0
D	A	A	1	0

State Assigned

PS	NS		O/P	
	x=0	x=1	x=0	x=1
00	01	00	0	0
01	01	10	0	0
10	01	11	0	0
11	00	00	1	0

State excitation

D_1 D_0		D_1 D_0		O/P	
x=0		x=1		x=0	x=1
01		00		0	0
01		10		0	0
01		11		0	0
00		00		1	0

D_0

w \ Q_1Q_0	00	01	11	10
0	1	1	0	1
1	0	0	0	1

$D_0 = \bar{w}\bar{Q}_1 + \bar{Q}_1\bar{Q}_0$

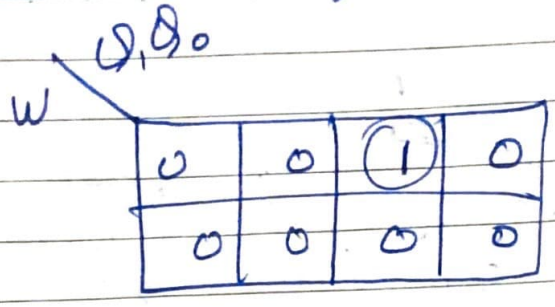
D_1

w \ Q_1Q_0	00	01	11	10
0	0	0	0	0
1	0	1	0	1

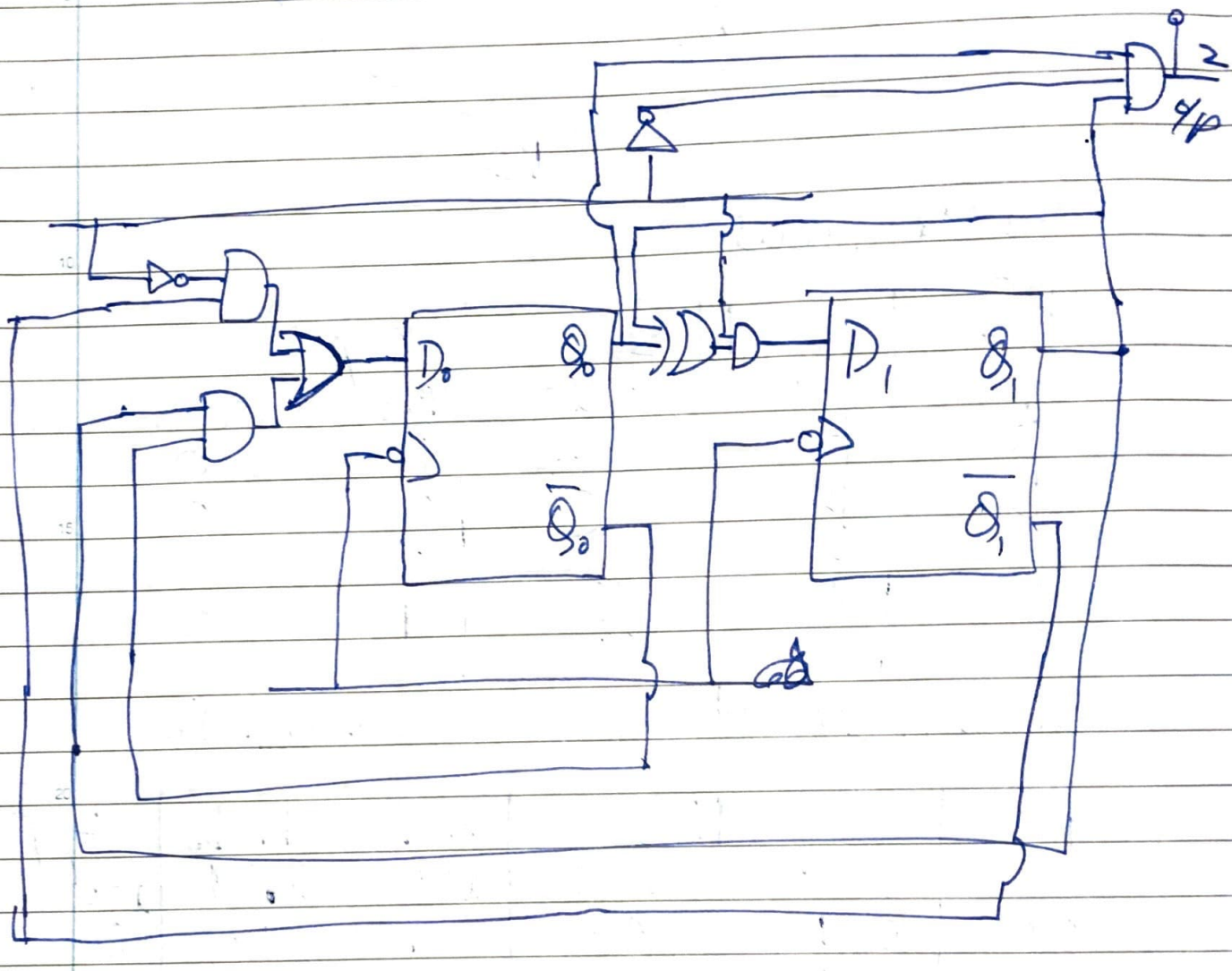
$D_1 = w\bar{Q}_1Q_0 + wQ_1\bar{Q}_0$

$= w(Q_1 \oplus Q_0)$

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$Z = \bar{w} Q_1 Q_0$



In overlapping the last bits of the detected sequence is allowed to act as the starting of next sequence whereas in non overlapping the it resets after the sequence is detected. eg 0110110

overlapping 0001001

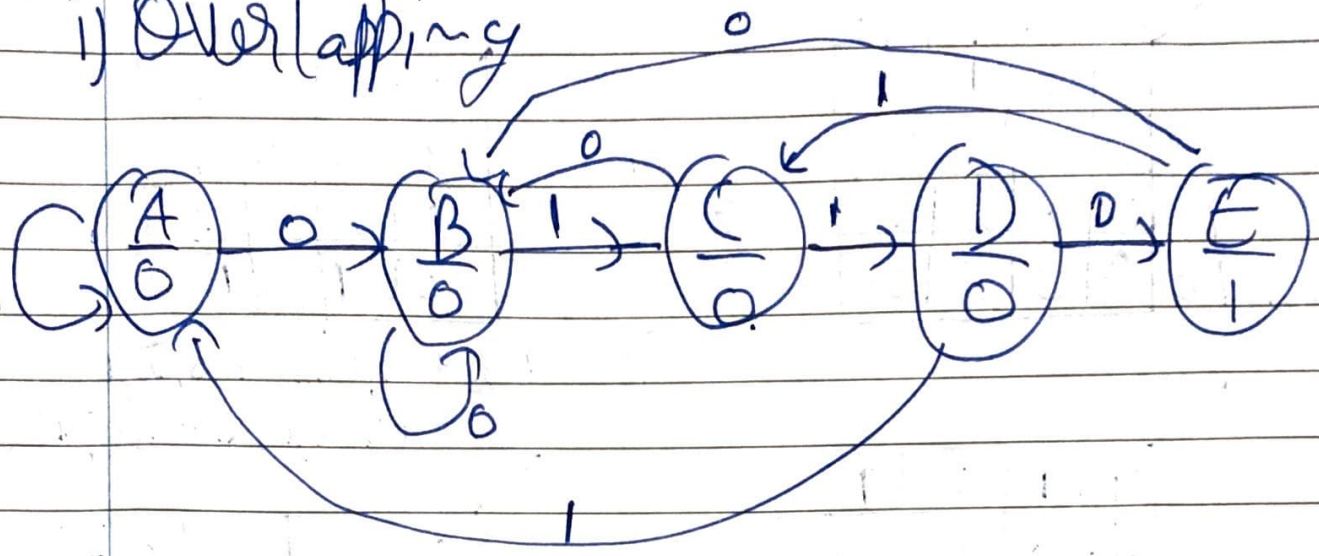
non " 0001000

In ~~appli~~ implementation overlapping is more complex

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24095 9318
Date :

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i) Overlapping



State Table

PS	NS x=0	NS x=1	O/P
A	B	A	0
B	B	C	0
C	B	D	0
D	E	A	0
E	B	C	1

State Assignment

PS	NS x=0	NS x=1
000	001	000
001	001	010
010	001	011
011	100	000
100	001	010

State Excitation

x=0	x=1	O/P
$D_2D_1D_0$	$D_2D_1D_0$	
001	000	0
001	010	0
001	011	0
100	000	0
001	010	1

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Date :

Q_1Q_0 \ Q_2X	00	01	11	10
00	0	0	1	0
01	0	X	X	X
11	0	X	X	X
10	0	0	0	0

$D_2 = Q_1\bar{Q}_0\bar{X}$

D_1

0	0	0	0
0	X	X	X
1	X	X	X
0	1	0	1

$D_1 = XQ_2 + \bar{Q}_1Q_0X + Q_1\bar{Q}_0X$

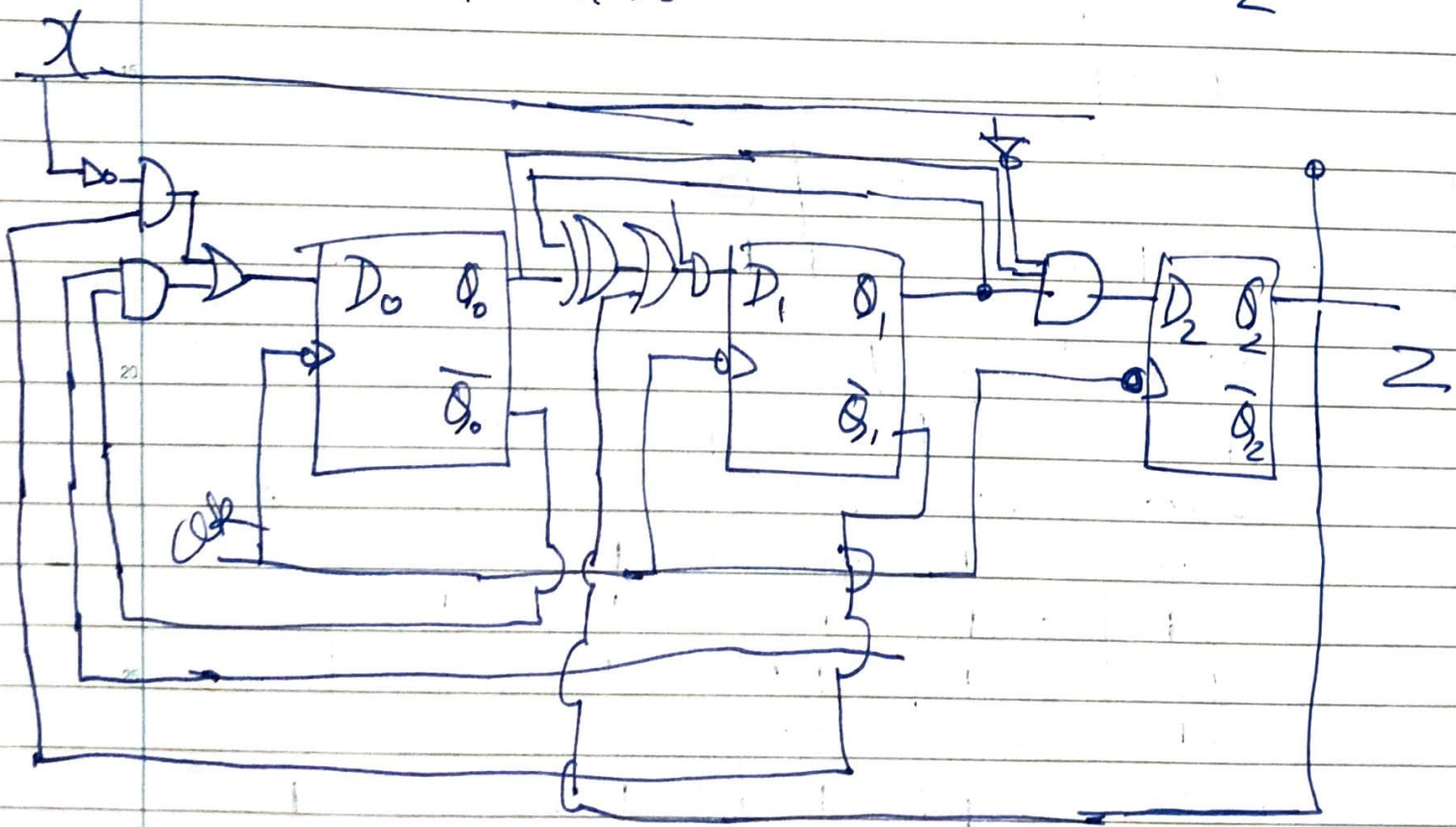
$= XQ_2 + X(Q_1 \oplus Q_0) = X(Q_2 + Q_1 \oplus Q_2)$

1	1	0	1
1	X	X	X
0	X	X	X
1	0	1	1

$D_0 = \bar{X}\bar{Q}_1 + Q_1\bar{Q}_0$

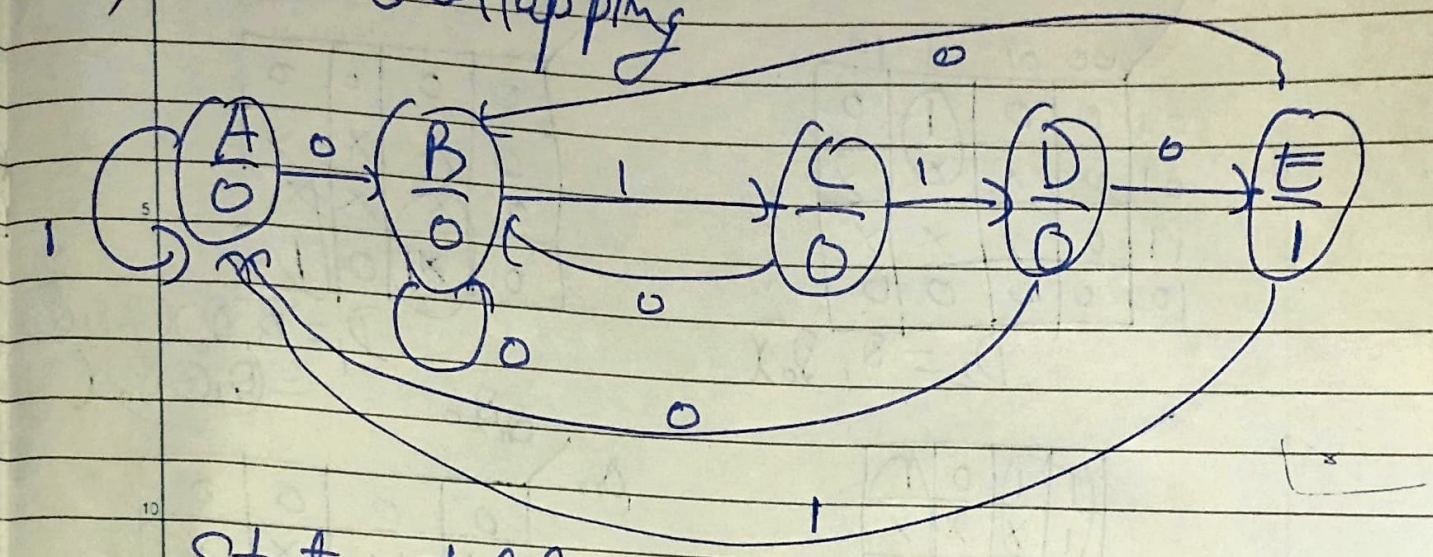
0	0	0	0
1	X	X	X

$Z = Q_2$



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(ii) Non Overlapping



State table

PS	NS		O/P
	x=0	x=1	
A	B	A	0
B	B	C	0
C	B	D	0
D	E	A	0
E	B	A	1

State Assignment

PS	NS	
~~000~~	x=0	x=1
000	001	000
001	001	010
010	001	011
011	100	000
100	001	000

State Excitation

$D_2D_1D_0$		O/P
x=0	x=1	
001	000	0
001	010	0
001	011	0
100	000	0
001	000	1

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Date :

Q_1Q_0 / X	00	01	11	10
00	0	0	1	0
01	0	x	x	x
11	0	x	x	x
10	0	0	0	0

$D_2 = Q_1 Q_0 X$

Q_1Q_0 / X	00	01	11	10
00	0	0	0	0
01	0	x	x	x
11	0	x	x	x
10	0	1	0	1

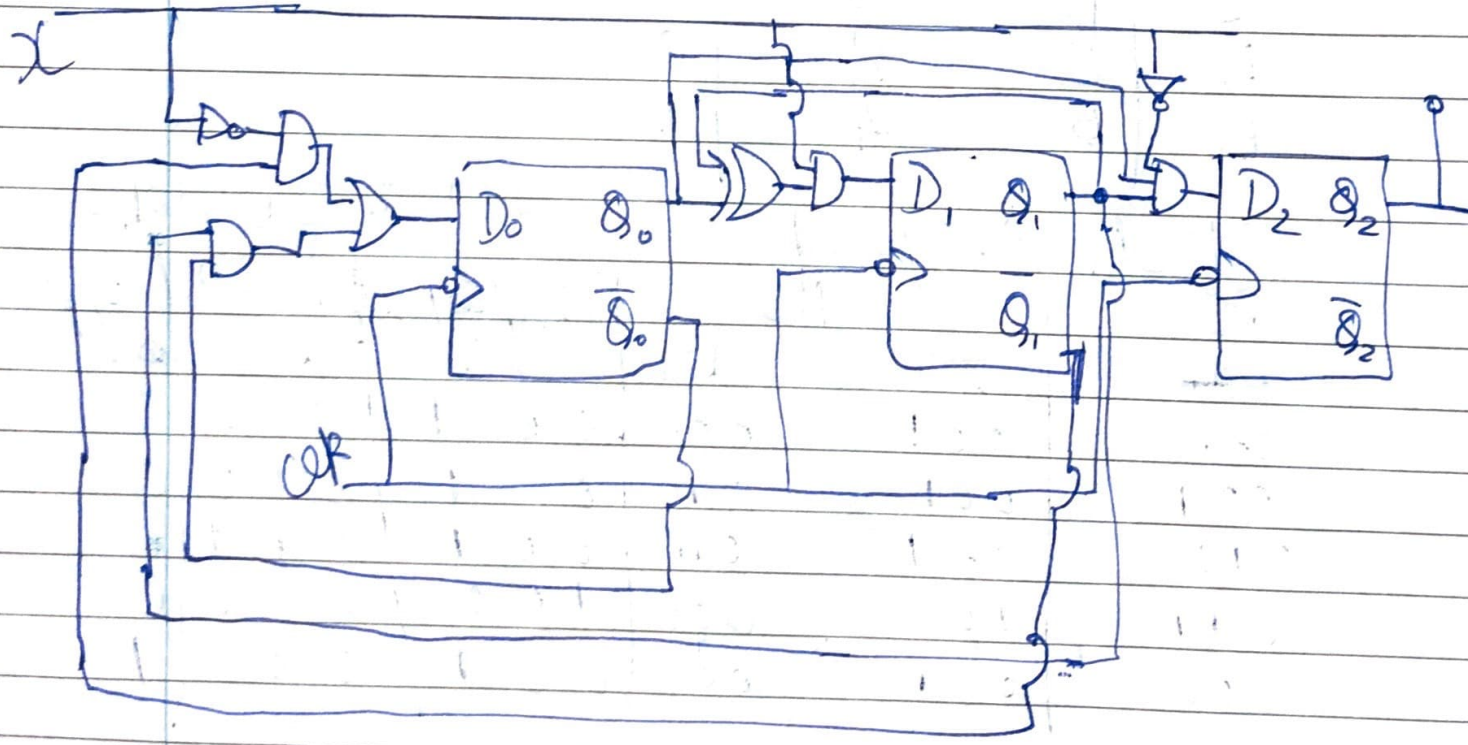
$D_1 = \bar{Q}_1 Q_0 X + Q_1 \bar{Q}_0 \bar{X}$
$= (Q_1 \oplus Q_0) X$

Q_1Q_0 / X	00	01	11	10
00	1	1	0	1
01	1	x	x	x
11	0	x	x	x
10	0	0	0	1

$D_0 = \bar{X} \bar{Q}_1 + \bar{Q}_1 \bar{Q}_0$

Q_2 / Q_1Q_0	00	01	11	10
0	0	0	0	0
1	1	x	x	x

$Z = Q_2$



Date : _____

Moore	Mealy
Output depends only on current state not on input	Output depends on both the current state & input
Output is associated with each state	Output changes immediately in response to input change even if state has not changed
Output = f(state)	Output = f(state, input)